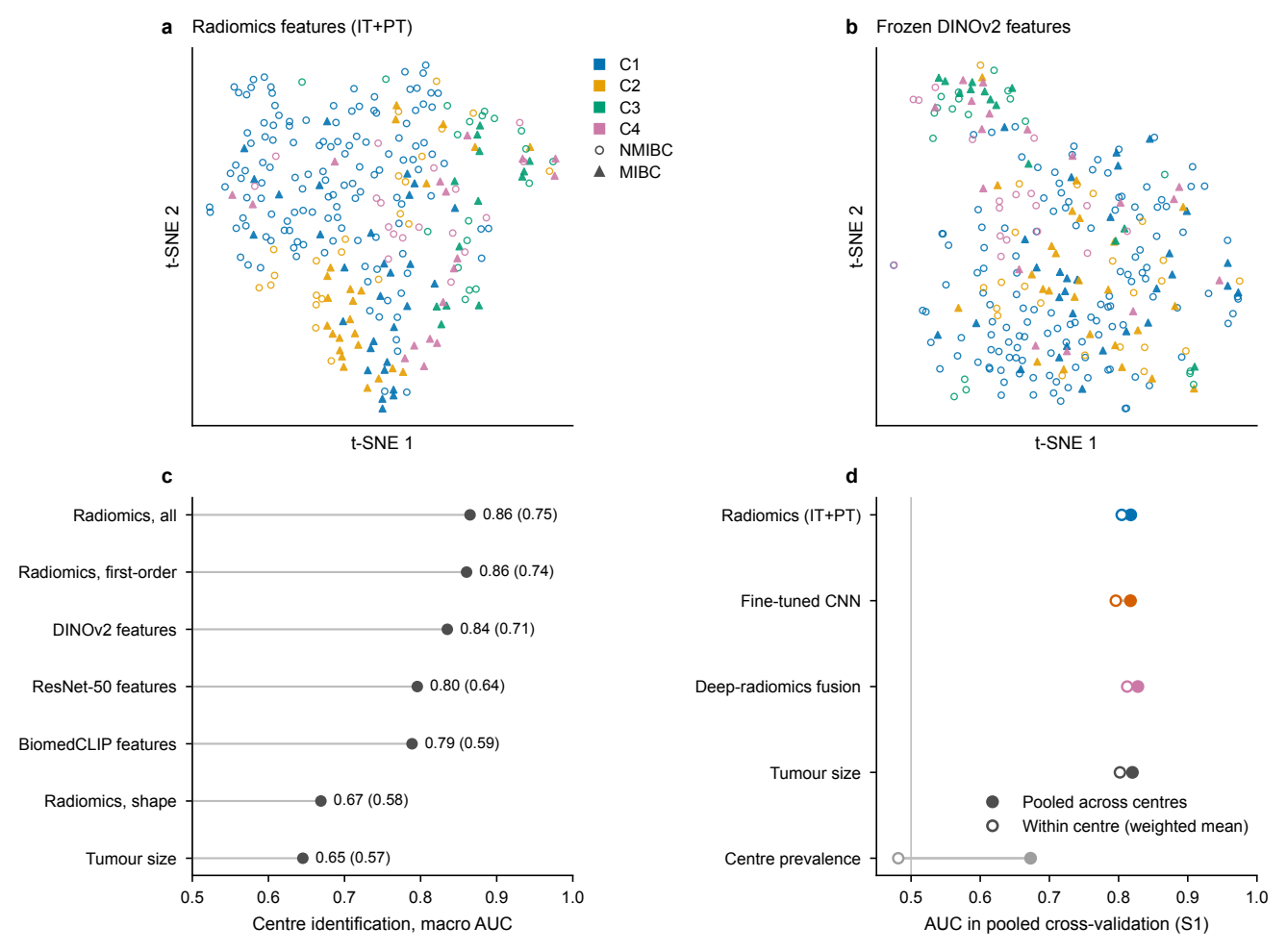
a
Radiomics features (IT+PT)
C1
C2
C3
C4
NMIBC
MIBC
t-SNE 2
t-SNE 1
b
Frozen DINOv2 features
t-SNE 2
t-SNE 1
c
Radiomics, all
0.86 (0.75)
Radiomics, first-order
0.86 (0.74)
DINOv2 features
0.84 (0.71)
ResNet-50 features
0.80 (0.64)
BiomedCLIP features
0.79 (0.59)
Radiomics, shape
0.67 (0.58)
Tumour size
0.65 (0.57)
0.5
0.6
0.7
0.8
0.9
1.0
Centre identification, macro AUC
d
Radiomics (IT+PT)
Fine-tuned CNN
Deep-radiomics fusion
Tumour size
Centre prevalence
Pooled across centres
Within centre (weighted mean)
0.5
0.6
0.7
0.8
0.9
1.0
AUC in pooled cross-validation (S1)